

COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
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MIDTERM EXAM Installation Procedure for Prometheus

1. PROCEDURE OUTPUT

Task 1: Prepare the Server

List the commands required to:

1. Update the package repository.
2. Upgrade installed packages.

```
GNU nano 6.2 main.yaml
- name: Prepare the Server (Ubuntu)
  ansible.builtin.apt:
    update_cache: yes
    upgrade: dist
  when: ansible_distribution == "Ubuntu"

- name: Prepare the Server (CentOS)
  ansible.builtin.apt:
    update_cache: yes
    upgrade: dist
  when: ansible_distribution == "CentOS"
```

3. Install the utilities required by the installation

- sudo apt install wget tar curl

```
- name: Install utilities required for installation (Ubuntu)
  ansible.builtin.apt:
    name:
      - wget
      - tar
      - curl
    state: present
  when: ansible_distribution == "Ubuntu"
```

```
- name: Install utilities required for installation (CentOS)
  ansible.builtin.dnf:
    name:
      - wget
      - tar
      - curl
    state: present
  when: ansible_distribution == "CentOS"
```

Task 2: Create the Prometheus User

```
- name: Create Prometheus Group
  ansible.builtin.group:
    name: prometheus
    state: present

- name: Create Prometheus User
  user:
    name: prometheus
    groups: prometheus
    append: yes
```

Part 3: Create Prometheus Directories

Identify the commands required to:

1. Create /etc/prometheus.
 - `sudo mkdir /etc/prometheus`
2. Create /var/lib/prometheus.
 - `sudo mkdir /var/lib/prometheus`
3. Assign the appropriate owner and group.
 - `sudo chown prometheus:prometheus /var/lib/prometheus`

```
- name: Create prometheus directories (etc/prometheus)
  ansible.builtin.file:
    path: /etc/prometheus
    state: directory

- name: Create prometheus directories (/var/lib/prometheus)
  ansible.builtin.file:
    path: /var/lib/prometheus
    owner: prometheus
    group: prometheus
    mode: '755'
    state: directory
```

Part 4: Download Prometheus

Identify the commands required to:

1. Navigate to a temporary working directory.
2. Download the Prometheus archive.
4. Extract the Prometheus archive.

```
- name: Download Prometheus
  ansible.builtin.get_url:
    url: https://github.com/prometheus/prometheus/releases/download/v2.55.1/prometheus-2.55.1-linux-amd64.tar.gz
    dest: ~/tmp

- name: Extract Prometheus
  ansible.builtin.unarchive:
    src: ~/tmp/prometheus-2.55.1-linux-amd64.tar.gz
    dest: ~/tmp
```

Task 5: Install the Prometheus Binaries

Identify the commands required to:

1. Locate the prometheus binary.
3. Copy both binaries to /usr/local/bin.
4. Set the appropriate ownership.
5. Verify the installed Prometheus version.
6. Verify the installed promtool version.

```
- name: Install Prometheus binaries (Prometheus)
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1-linux-amd64/prometheus
    dest: /usr/local/bin/
    owner: prometheus
    group: prometheus

- name: Install Prometheus binaries (promtool)
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1-linux-amd64/promtool
    dest: /usr/local/bin/
    owner: prometheus
    group: prometheus
```

Task 6: Install the Prometheus Configuration Files

Identify the commands required to:

1. Copy prometheus.yml to /etc/prometheus.
sudo cp /tmp/prometheus-2.55.1-linux-amd64/prometheus.yml /etc/prometheus/
2. Copy the consoles directory.
sudo cp -r /tmp/prometheus-2.55.1-linux-amd64/consoles /etc/prometheus/
3. Copy the console_libraries directory.
sudo cp -r /tmp/prometheus-2.55.1-linux-amd64/console_libraries /etc/prometheus/
4. Set the appropriate ownership.
sudo chown -R prometheus:prometheus /etc/prometheus
5. Verify the files and directories.

```
- name: Install the prometheus Configuration Files
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/prometheus.yml
    dest: /etc/prometheus/
    owner: prometheus
    group: prometheus

- name: Install prometheus console files
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/consoles
    dest: /etc/prometheus/
    owner: prometheus
    group: prometheus

- name: Install prometheus console library files
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/console_libraries
    dest: /etc/prometheus/
    owner: prometheus
    group: prometheus
```

```
TASK [prometheus : Create Prometheus User] *****
ok: [server2]

TASK [prometheus : Create prometheus directories (etc/prometheus)] *****
ok: [server2]

TASK [prometheus : Create prometheus directories (/var/lib/prometheus)] *****
ok: [server2]

TASK [prometheus : Download Prometheus] *****
ok: [server2]

TASK [prometheus : Extract Prometheus] *****
ok: [server2]

TASK [prometheus : Install Prometheus binaries (Prometheus)] *****
ok: [server2]

TASK [prometheus : Install Prometheus binaries (promtool)] *****
ok: [server2]

TASK [prometheus : Install the prometheus Configuration Files] *****
changed: [server2]

TASK [prometheus : Install prometheus console files] *****
changed: [server2]

TASK [prometheus : Install prometheus console library files] *****
changed: [server2]

PLAY RECAP *****
server1 : ok=1   changed=0   unreachable=0   failed=1   skipped=0   rescued=0   ignored=0
server2 : ok=13  changed=3   unreachable=0   failed=0   skipped=1   rescued=0   ignored=0
```

2. PLAYBOOK SOURCE CODE

My ubuntu server wasn't working. I tried replacing it with other ubuntu servers. Cloning. Changing IP but it wouldn't do update properly if not

```
#- name: Prepare the Server (Ubuntu)
#  ansible.builtin.apt:
#    update_cache: yes
#    state: latest
#  when: ansible_distribution == "Ubuntu"

#- name: Prepare the Server (CentOS)
#  ansible.builtin.dnf:
#    update_cache: yes
#    state: latest
#  when: ansible_distribution == "CentOS"

- name: Install utilities required for installation (Ubuntu)
  ansible.builtin.apt:
    name:
      - wget
      - tar
      - curl
    state: present
  when: ansible_distribution == "Ubuntu"

- name: Install utilities required for installation (CentOS)
  ansible.builtin.dnf:
    name:
      - wget
      - tar
      - curl
    state: present
  when: ansible_distribution == "CentOS"

ansible.builtin.group:
  name: prometheus
  state: present

- name: Create Prometheus User
  user:
    name: prometheus
    groups: prometheus
    append: yes

- name: Create prometheus directories (etc/prometheus)
  ansible.builtin.file:
    path: /etc/prometheus
    state: directory

- name: Create prometheus directories (/var/lib/prometheus)
  ansible.builtin.file:
    path: /var/lib/prometheus
    owner: prometheus
    group: prometheus
    mode: '755'
    state: directory

- name: Download Prometheus
  ansible.builtin.get_url:
    url: https://github.com/prometheus/prometheus/releases/download/v2.55.1/prometheus-2.55.1.linux-amd64.tar.gz
    dest: /tmp

- name: Extract Prometheus
  ansible.builtin.unarchive:
    src: /tmp/prometheus-2.55.1.linux-amd64.tar.gz
    remote_src: yes
    dest: /tmp
```

```
- name: Install Prometheus binaries (promtool)
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/promtool
    dest: /usr/local/bin/
    remote_src: yes
    owner: prometheus
    group: prometheus

- name: Install the prometheus Configuration Files
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/prometheus.yml
    dest: /etc/prometheus/
    remote_src: yes
    owner: prometheus
    group: prometheus

- name: Install prometheus console files
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/consoles
    dest: /etc/prometheus/
    remote_src: yes
    owner: prometheus
    group: prometheus

- name: Install prometheus console library files
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/console_libraries
    dest: /etc/prometheus/
    remote_src: yes
    owner: prometheus
    group: prometheus
```

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

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Extent of Use:

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